WORKED SOLUTION

STEP III 2023, Question 10

Mechanics • Equilibrium of a Beam

STEP III 2023, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

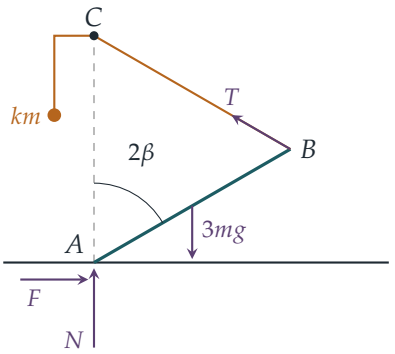
www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The pulley sits exactly a beam-length above A, so A, B and C form an isosceles triangle with $AB = AC = 2h$. That geometry hands us the direction of the string and its perpendicular distance from A for free. Then everything follows from the usual recipe for a rod in equilibrium: take moments about A (which removes both unknown contact forces), resolve, and impose $F \leq \mu N$.

Solution

We model the beam as a uniform rod, the particles as point masses, and the string as light and inextensible over a smooth pulley, so the tension is the same throughout. Since the hanging particle is in equilibrium, the tension is $T = kmg$.



Put A at the origin with the x-axis horizontal (towards the side of B) and the y-axis vertically up. Then $C = (0, 2h)$ and $B = 2h(\sin 2\beta, \cos 2\beta)$.

Direction of the string. The vector from B to C is

$$2h\big(-\sin 2\beta,\ 1-\cos 2\beta\big) = 4h\sin\beta\,(-\cos\beta,\ \sin\beta),$$

using $1 - \cos 2\beta = 2\sin^2\beta$. So the tension on the beam has horizontal component $T\cos\beta$ in the $-x$ direction (back towards the line AC) and vertical component $T\sin\beta$ upwards. Geometrically: in the isosceles triangle ABC the base angles are $90^\circ - \beta$, so BC makes angle β with the horizontal.

Moment arm. The perpendicular from A to BC bisects the apex angle 2β of the isosceles triangle, so its length is $2h\cos\beta$.

Part (i)

Moments about A. The weight $3mg$ acts at the midpoint of AB, a horizontal distance $h\sin 2\beta$ from A. Balancing it against the tension:

$$kmg \cdot 2h\cos\beta = 3mg \cdot h\sin 2\beta = 6mgh\sin\beta\cos\beta.$$

Since $0 < 2\beta < 180^\circ$, $\cos\beta \neq 0$, and we may divide by it:

$$\boxed{k = 3\sin\beta.}$$

(This is consistent with $k < 3$.)

Resolving. Vertically, $N + T\sin\beta = 3mg$, so

$$N = 3mg - 3mg\sin^2\beta = 3mg\cos^2\beta.$$

Horizontally, friction must balance the horizontal pull of the string, so it acts in the $+x$ direction with

$$F = T\cos\beta = 3mg\sin\beta\cos\beta.$$

Hence $F/N = \tan\beta$.

Friction condition. Equilibrium requires $F \leq \mu N$, that is $\tan\beta \leq \mu$. Now

$$k^2 = 9\sin^2\beta = \frac{9\tan^2\beta}{1+\tan^2\beta}.$$

The function $t \mapsto \dfrac{t}{1+t} = 1 - \dfrac{1}{1+t}$ is increasing for $t \geq 0$, and $0 \leq \tan^2\beta \leq \mu^2$, so

$$k^2 \leq \frac{9\mu^2}{\mu^2+1},$$

as required.

Part (ii)

Now $T = 2mg$, and the extra particle of mass m sits a horizontal distance $xh\sin 2\beta$ from A. The geometry of the string is unchanged.

Moments about A:

$$2mg \cdot 2h\cos\beta = 3mg\,h\sin 2\beta + mg\,xh\sin 2\beta = 2mgh(3+x)\sin\beta\cos\beta,$$

and dividing by $2mgh\cos\beta \neq 0$ gives

$$\sin\beta = \frac{2}{3+x}.$$

For $0 \leq x \leq 2$ this lies between $\frac{2}{5}$ and $\frac{2}{3}$, so such a β exists.

Resolving:

$$N = 4mg - 2mg\sin\beta, \qquad F = 2mg\cos\beta,$$

so

$$\frac{F^2}{N^2} = \frac{4\cos^2\beta}{(4-2\sin\beta)^2} = \frac{\cos^2\beta}{(2-\sin\beta)^2}.$$

Substituting $\sin\beta = 2/(3+x)$:

$$\cos^2\beta = 1 - \frac{4}{(3+x)^2} = \frac{(3+x-2)(3+x+2)}{(3+x)^2} = \frac{(x+1)(x+5)}{(3+x)^2},$$
$$2-\sin\beta = \frac{2(3+x)-2}{3+x} = \frac{2(x+2)}{3+x}.$$

The factors of $(3+x)^2$ cancel, leaving

$$\frac{F^2}{N^2} = \frac{(x+1)(x+5)}{4(x+2)^2} = \frac{x^2+6x+5}{4(x+2)^2},$$

which is the required result.

Ask yourself How do we turn "whatever the value of x" into a condition on μ?

For each x we need $F \leq \mu N$, i.e. $\mu \geq F/N$. To cover every $x \in [0,2]$, μ must be at least the *largest* value of F/N over that interval. So we want the maximum of F^2/N^2.

Following the hint,

$$\frac{1}{3} - \frac{F^2}{N^2} = \frac{4(x+2)^2 - 3(x^2+6x+5)}{12(x+2)^2} = \frac{x^2-2x+1}{12(x+2)^2} = \frac{(x-1)^2}{12(x+2)^2} \geq 0,$$

with equality exactly when $x = 1$, which is in the allowed range. So $F^2/N^2 \leq \frac{1}{3}$ for all $x \in [0,2]$, and the bound is attained at $x = 1$. Hence F/N has maximum value $1/\sqrt{3}$, and the beam can be in equilibrium for every x if and only if $\mu \geq 1/\sqrt{3}$. The minimum value is

$$\boxed{\mu = \frac{1}{\sqrt{3}}}.$$

Checks

- In part (i), try the beam lying horizontal, $2\beta = 90°$. Then $B = (2h, 0)$, the string BC is at $45°$ and its distance from A is $\sqrt{2}\,h$. Moments give $T\sqrt{2}\,h = 3mgh$, so $T = 3mg/\sqrt{2}$, matching $k = 3\sin 45°$. Also $N = F = \frac{3}{2}mg$, so $F/N = 1 = \tan 45°$.

- At $x = 0$ in part (ii) the formula gives $5/16$, and directly $\sin\beta = \frac{2}{3}$, $\cos^2\beta = \frac{5}{9}$, $(2 - \frac{2}{3})^2 = \frac{16}{9}$, giving $\frac{5}{16}$ again.
- At $x = 1$: $\sin\beta = \frac{1}{2}$, so $\beta = 30^\circ$ and the beam is at 60° to the vertical; $F/N = \cos 30^\circ / (2 - \frac{1}{2}) = 1/\sqrt{3} \approx 0.5774$. A computer check of the moments and resolved forces for x across $[0, 2]$ confirmed the formula and this maximum.

Common slips

- Forgetting that the string pulls along BC, not vertically: the pulley is at C, so the tension has a horizontal component, and that is what friction has to resist.
- Taking the minimum of F/N (or its value at an end point) instead of its maximum over $0 \le x \le 2$. The worst case is in the interior, at $x = 1$.
- Mixing up the moment arms: the weight's arm is horizontal ($h\sin 2\beta$), while the tension's arm is the perpendicular distance to BC ($2h\cos\beta$).

Beyond STEP

Taking moments and resolving are the two halves of the equilibrium conditions for a rigid body: zero total force and zero total torque. At university these become vector statements, $\sum \mathbf{F} = \mathbf{0}$ and $\sum \mathbf{r} \times \mathbf{F} = \mathbf{0}$, and the "worst case over a parameter" reasoning of part (ii) is a small taste of how engineers design for safety margins. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops statics and torque carefully in this vector language.

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